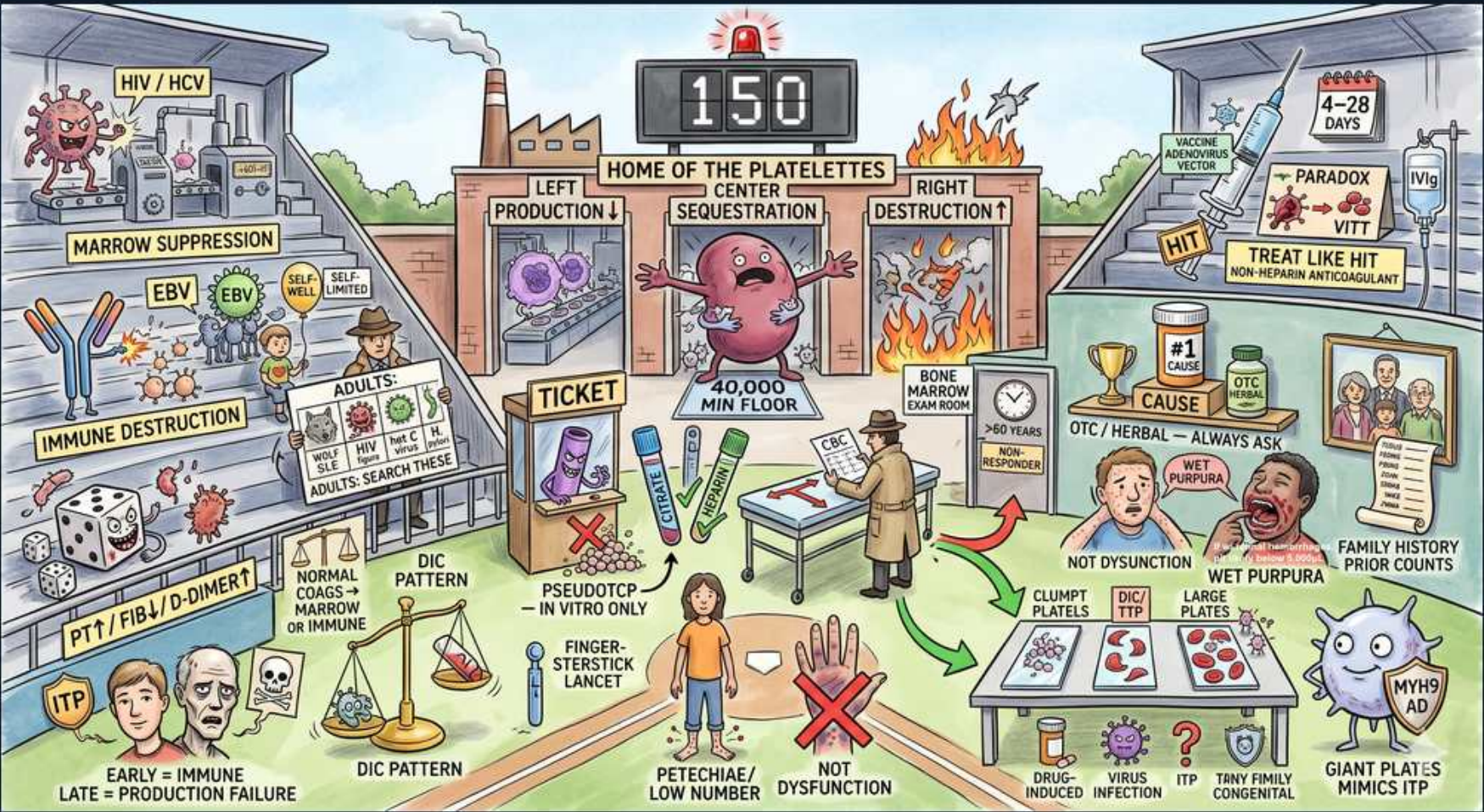


Platelet disorders — thrombocytopenia approach (Home of the Platelettes)



1. Threshold 150K

WHAT YOU SEE

A giant red stadium scoreboard reading 150 above 'HOME OF THE PLATELETTES' — the lit number is the 150,000/uL cutoff that defines thrombocytopenia.

WHAT IT ENCODES

Thrombocytopenia is defined as a platelet count $<150,000/\mu\text{L}$ (normal 150–450K). The number alone does not predict bleeding linearly — spontaneous mucocutaneous bleeding is uncommon until $<10\text{--}20\text{K}$, and most invasive procedures/surgery can proceed safely above 50K. Counts of 100–150K are often incidental and benign (e.g., gestational), so always anchor the workup to the trend and the smear, not a single value.

BOARD TRAP

Don't transfuse platelets just because the count is 'low' (e.g., 80K) in a stable, non-bleeding patient — the threshold for prophylactic transfusion is generally $<10\text{K}$ (or $<20\text{K}$ with fever/sepsis, $<50\text{K}$ for surgery/bleeding). Reflexively transfusing wastes product and is actively harmful in TTP/HIT (fuels thrombosis).

2. Three mechanisms

WHAT YOU SEE

The platelet factory split into three labeled bays — PRODUCTION (down-arrow), SEQUESTRATION, and DESTRUCTION (in flames) — the three buckets every low count must fall into.

WHAT IT ENCODES

Every thrombocytopenia sorts into one of three buckets: decreased PRODUCTION (marrow problem), splenic SEQUESTRATION (redistribution), or increased DESTRUCTION/consumption (peripheral). A fourth, dilutional, occurs after massive transfusion/fluids. The fastest way to separate them is the smear plus an MPV/reticulated platelet check — destruction and sequestration give young, large platelets, while production failure gives few small platelets often with other cytopenias.

BOARD TRAP

Don't assume isolated thrombocytopenia means destruction — pancytopenia or a leukoerythroblastic smear points to a marrow (production) process and mandates marrow exam, not a course of steroids for presumed ITP.

3. Decreased production

WHAT YOU SEE

The left factory bay with its conveyor slowed and the 'PRODUCTION down' sign — the stalled assembly line is the marrow failing to make platelets.

WHAT IT ENCODES

Production failure comes from marrow suppression (chemo, radiation, alcohol, viral — HIV/HCV/EBV/CMV/parvovirus), marrow infiltration/replacement (leukemia, lymphoma, myelofibrosis, metastatic cancer → leukoerythroblastic smear with teardrops and nucleated RBCs), or nutritional deficiency (B12/folate causing ineffective megaloblastic hematopoiesis, hypersegmented neutrophils). Typically OTHER cell lines are also low. Reticulated/immature platelet fraction is LOW because the marrow is not producing.

BOARD TRAP

Don't call low platelets + macro-ovalocytes + hypersegmented neutrophils 'ITP' — that pattern is B12/folate deficiency; treat the vitamin, not with immunosuppression.

4. Sequestration

WHAT YOU SEE

The huge spleen swallowing platelets with a '40,000 MIN FLOOR' tag — the spleen hoarding the platelets, and the floor sign says sequestration alone rarely drops below 40K.

WHAT IT ENCODES

A normal spleen pools $\sim 1/3$ of platelets; with splenomegaly (cirrhosis/portal hypertension, lymphoma, Gaucher, infiltration) up to 90% are sequestered. The hallmark is a moderate floor — sequestration alone rarely drops counts below $\sim 40,000/\mu\text{L}$, and total platelet mass/lifespan is normal, so bleeding is mild. Look for the cause of splenomegaly (chronic liver disease is #1 in adults) rather than treating the count.

BOARD TRAP

Don't attribute a platelet count of 8K purely to hypersplenism — sequestration floors around 40K; a count that low means a concurrent destruction or production process is present.

5. Destruction

WHAT YOU SEE

The right factory bay engulfed in flames — the fire is peripheral destruction/consumption burning platelets up faster than they are made.

WHAT IT ENCODES

Peripheral destruction/consumption is the 'fire' bucket: immune (ITP, drug-induced including HIT) and non-immune microangiopathic (TTP, HUS, DIC). Smear is the discriminator — schistocytes signal MAHA (TTP/HUS/DIC) and demand urgent action, while a clean smear with only large platelets suggests ITP. The reticulated platelet fraction is HIGH because the marrow is compensating for rapid turnover.

BOARD TRAP

Don't lump all destructive thrombocytopenias together — platelet transfusion helps in ITP bleeding but is contraindicated in TTP and HIT, where it adds substrate for thrombosis. Always check a smear and coags before transfusing.

6. Pseudothrombocytopenia

WHAT YOU SEE

A ticket/clump stub labeled 'PSEUDO-TCP — IN VITRO' showing platelets stuck together — the in-vitro EDTA clumping that tricks the analyzer into a falsely low count.

WHAT IT ENCODES

Pseudothrombocytopenia is an in-vitro artifact: EDTA in the purple-top tube triggers naturally occurring antibodies to clump platelets, so the automated analyzer counts clumps as single cells and reports a falsely low number. The patient is asymptomatic with no bleeding. Confirm by examining the smear (visible clumps) and redrawing in a citrate (blue-top) or heparin tube — the count normalizes.

BOARD TRAP

Don't work up or treat a clumping artifact — repeat in citrate first. Starting steroids, IVIG, or transfusion for a falsely low EDTA count is a classic harm. The tip-off: 'asymptomatic, isolated severe thrombocytopenia, no bleeding' on a routine CBC.

7. Smear review first

WHAT YOU SEE

A trench-coated detective peering at a blood smear through a magnifier — the smear is the cheap first investigation that cracks every thrombocytopenia case.

WHAT IT ENCODES

The peripheral smear is the essential, cheap first step in any thrombocytopenia. It rules in/out pseudothrombocytopenia (clumps), MAHA (schistocytes), acute leukemia (blasts), TTP/DIC, and inherited macrothrombocytopenia (giant platelets). It also lets you eyeball the count ($\approx$ platelets per oil field $\times$ 15,000) and check the other two cell lines. No diagnostic algorithm should advance until the smear is seen.

BOARD TRAP

Don't order ADAMTS13, marrow biopsy, or immune workups before looking at the smear — skipping the smear is the classic way to miss schistocytes (TTP, a medical emergency) or platelet clumps (a non-disease).

8. Schistocytes → MAHA

WHAT YOU SEE

Smear with fragmented cells

WHAT IT ENCODES

Schistocytes (fragmented, helmet-shaped RBCs) on the smear define microangiopathic hemolytic anemia (MAHA): RBCs sheared as they pass platelet/fibrin microthrombi. MAHA + thrombocytopenia is a thrombotic microangiopathy until proven otherwise — TTP, HUS, or DIC. Confirm hemolysis with high LDH, high indirect bilirubin, low haptoglobin, and negative direct Coombs (non-immune). Send coags and ADAMTS13 immediately; if TTP is plausible, start plasma exchange empirically.

BOARD TRAP

Don't wait for the ADAMTS13 result before treating suspected TTP — mortality is $\sim$ 90% untreated. Empiric plasma exchange (plus steroids $\pm$ caplacizumab) is started on the clinical triad of MAHA + thrombocytopenia without another explanation.

9. DIC pattern

WHAT YOU SEE

A pair of dice labeled DIC beside a scale of coags (PT/PTT/D-dimer/fibrinogen) — the rolled-dice chaos of consumptive coagulopathy with its prolonged-coags, high-D-dimer, low-fibrinogen signature.

WHAT IT ENCODES

DIC is consumptive coagulopathy from a systemic trigger (sepsis, malignancy especially APL, trauma, obstetric catastrophe). The lab signature is PROLONGED PT and aPTT, ELEVATED D-dimer (and FDPs), and LOW fibrinogen, with falling platelets and schistocytes — this coagulation derangement is exactly what distinguishes DIC from TTP/HUS, where PT/aPTT and fibrinogen are normal. Treat the underlying cause; replace with FFP/cryoprecipitate/platelets only if bleeding.

BOARD TRAP

Normal coags rule out DIC — reconsider TTP/HUS. Conversely, treating DIC with anticoagulation or factor replacement without addressing the trigger (e.g., source control in sepsis, ATRA in APL) fails — DIC is a symptom of an underlying catastrophe, not a primary diagnosis.

10. Always ask drugs/OTC

WHAT YOU SEE

An open pill bottle tagged 'ALWAYS ASK — OTC/HERBAL' — the reminder that the drug list (including tonic-water quinine and heparin flushes) is mandatory and easily missed.

WHAT IT ENCODES

Drug history is mandatory and frequently missed — include prescription, OTC, herbals, supplements, quinine (tonic water!), and especially heparin (any form, including flushes). Drug-induced immune thrombocytopenia typically appears $\sim$ 1–2 weeks after a new drug (faster on re-exposure) and resolves 7–10 days after stopping. Classic culprits: heparin, quinine/quinidine, sulfa drugs, vancomycin, GPIIb/IIIa inhibitors, valproate, linezolid.

BOARD TRAP

Don't anchor on ITP and skip the medication list — a missed quinine (tonic water) or heparin exposure means the real fix (stop the drug) is delayed and immunosuppression is started unnecessarily.

11. Marrow exam if unexplained

WHAT YOU SEE

A 'BONE MARROW EXAM ROOM' door — the marrow biopsy reserved for unexplained thrombocytopenia with cytopenias, blasts, or age >60 .

WHAT IT ENCODES

Bone marrow biopsy/aspirate is reserved for thrombocytopenia with features suggesting a production problem: additional cytopenias, abnormal cells (blasts) on smear, organomegaly/lymphadenopathy, age >60 with new isolated thrombocytopenia, or failure to respond to ITP therapy. Megakaryocytes are increased in destruction (ITP) and decreased/abnormal in production failure (aplasia, MDS, infiltration).

BOARD TRAP

Don't biopsy the marrow for classic isolated ITP in a young patient — it's unnecessary; but DO biopsy before committing an older patient with 'ITP' to chronic immunosuppression, since MDS can masquerade as isolated thrombocytopenia.

12. Petechiae = low number

WHAT YOU SEE

A hand stippled with pinpoint petechiae labeled 'NOT DYSFUNCTION' — petechiae/mucosal bleeding signal a low platelet NUMBER (primary hemostasis), not a clotting-factor problem.

WHAT IT ENCODES

Bleeding pattern localizes the lesion. Petechiae, purpura, mucosal/gingival bleeding, and epistaxis reflect a QUANTITATIVE platelet problem (low number) or qualitative platelet dysfunction — i.e., primary hemostasis. By contrast, deep bleeds, hemarthroses, and delayed bleeding point to a coagulation factor (secondary hemostasis) defect. Petechiae appearing only when count is very low (<10 – 20 K) reflects number, not platelet function.

BOARD TRAP

Don't interpret petechiae as a clotting-factor problem — petechiae/mucosal bleeding = platelet (primary hemostasis); hemarthrosis and deep muscle bleeds = factor deficiency (hemophilia). Mixing these up sends the workup down the wrong pathway.

13. ITP = diagnosis of exclusion

WHAT YOU SEE

An ITP locker tagged 'EARLY = IMMUNE, LATE = PRODUCTION FAILURE' — the double-hit immune destruction plus impaired production, diagnosed only after excluding everything else.

WHAT IT ENCODES

ITP is immune-mediated platelet destruction (anti-GPIIb/IIIa antibodies) AND impaired production, diagnosed by ISOLATED thrombocytopenia with a normal smear after excluding drugs, infection (HIV, HCV, H. pylori), and other causes — there is no confirmatory test. First-line therapy is corticosteroids (prednisone or dexamethasone) ± IVIG (1 g/kg) for rapid rise or active bleeding; second-line includes TPO agonists (eltrombopag, romiplostim), rituximab, or splenectomy.

BOARD TRAP

Don't diagnose ITP without checking HIV and HCV — both classically present as 'ITP' and are reversible/treatable causes; immunosuppressing an undiagnosed HIV patient is the trap. Also test before steroids, since steroids can mask/worsen an underlying infection or lymphoma.

14. Giant platelets mimic ITP

WHAT YOU SEE

An oversized platelet bigger than a red cell labeled 'MYH9 / GIANT PLATELETS MIMIC ITP' — the inherited macrothrombocytopenia masquerading as ITP.

WHAT IT ENCODES

Inherited macrothrombocytopenias (MYH9-related disorders, Bernard-Soulier) produce giant platelets (larger than RBCs, high MPV) and a chronically low count that mimics ITP. Clues to inherited disease: lifelong low counts (check old CBCs), family history, syndromic features (MYH9: Döhle-like leukocyte inclusions, nephritis, deafness, cataracts), and NO response to steroids/IVIG/splenectomy. Bernard-Soulier is a GPIb deficiency causing giant platelets + bleeding.

BOARD TRAP

Don't immunosuppress an inherited giant-platelet disorder as if it were ITP — these patients won't respond to steroids/IVIG and shouldn't have a splenectomy. The discriminator is reviewing PRIOR platelet counts (lifelong low) and family history; chronic, treatment-refractory 'ITP' should prompt this reconsideration.
